

Lightweight Feature Selection for IDK Cascade Skip Decisions

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Abstract—IDK cascades chain progressively heavier classifiers so easy inputs exit early. Two recent improvements skip the middle classifier when the first stage is unlikely to be refined: Nguyen et al. (RTSS 2024) used a static confidence threshold; Katikaneni et al. (RTSS 2025) replaced it with a Random Forest on three softmax features. We ask whether a much simpler predictor suffices. Under Katikaneni et al.’s cross-dataset ImageNet-V2 protocol and prediction target (will the middle classifier abstain?), a two-feature logistic regression on *confidence* and *margin* matches the Random Forest on the first-stage-IDK test population: 365.3 vs. 363.8 ms/image at 55.3% vs. 55.3% top-1 accuracy on a Jetson Orin Nano CPU. The paired bootstrap 95% CI of the per-image accuracy difference includes zero; the latency CI shows the Random Forest ahead by at most 2.6 ms — less than its own 3.8 ms median per-decision evaluation cost, which the logistic undercuts by a measured $205\times$ (19 μ s at batch size 1). Unlike the tree ensemble, the deployed logistic is branch-free and data-independent, so its WCET is immediate to bound — the operative property where the skip predictor’s WCET itself enters the schedulability budget.

Index Terms—real-time machine learning, IDK cascades, any-time classification, edge inference, schedulability

I. INTRODUCTION

Modern real-time systems increasingly rely on deep neural networks for perception, control, and classification at the edge. The dominant tension in this setting is well known: more accurate networks have larger and more variable inference times, which makes them difficult to schedule under hard deadlines. *IDK cascades*, introduced by Wang et al. [1] and later formalized for real-time schedulability analysis by Baruah et al. [2], address this tension by chaining a sequence of classifiers $A \rightarrow B \rightarrow C$ of increasing accuracy and cost. At each stage, the classifier either commits to a prediction or returns “IDK” (I-Don’t-Know), in which case the next, heavier classifier is invoked. Easy inputs exit at A ; only difficult inputs incur the full $A + B + C$ cost.

A recurring observation in the IDK-cascade literature is that the middle classifier B is often *wasted execution*: when A already returns IDK with very low confidence, B is unlikely to do better, and the cascade should jump directly to the deterministic final classifier C . Nguyen, Cheng, and Carroll [3] formalized this observation with a probabilistic profiling phase and a single static threshold (skip B when A ’s top-1 confidence falls below 0.3), reporting 11.5–17.0% average latency reduction across three ImageNet-V2 subsets with no loss

of accuracy. Katikaneni, Cheng, and Carroll [4] argued that a single static threshold does not generalize across datasets, and replaced it with a Random Forest trained on three features of A ’s softmax distribution: *confidence*, *entropy*, and *margin*. They showed that the learned skip predictor outperforms the static threshold on three ImageNet-V2 subsets and on a Jetson Nano embedded target.

The Random Forest, however, introduces its own per-image inference cost. The 2025 paper itself flags as future work the question of whether a simpler tabular model would suffice. In this work-in-progress, we take that question literally: *is a two-feature logistic skip rule sufficient to recover most of the latency benefit reported in [4], while remaining interpretable and trivially cheap to evaluate at inference time?*

Contributions. (i) We reproduce the dynamic-threshold cascade of [3] with off-the-shelf pretrained ResNet-18/34/152 weights on ImageNet-V2. (ii) We implement a logistic-regression skip predictor over $\{\textit{confidence}, \textit{margin}\}$ and compare it to both the static threshold and a re-implementation of the Random Forest baseline with the same hyperparameters, under the cross-dataset protocol of [4] (Sec. IV), with bootstrap interval estimates and a nested-feature ablation. (iii) We measure the per-decision overhead of the skip predictor itself on an embedded ARM target and argue that, beyond its measured cost, the logistic’s branch-free evaluation is the more analyzable term to carry into a schedulability bound.

II. BACKGROUND AND RELATED WORK

A. IDK Cascades

A formal IDK cascade [2] is a sequence of classifiers $C_1, \dots, C_n$ ordered by non-decreasing average execution time $\bar{e}_i$ and accuracy a_i ; each C_i with $i < n$ may abstain by emitting an “IDK” token $\perp$, and C_n must always commit. Let $p_i \in [0, 1]$ denote the probability that C_i emits $\perp$ on a randomly drawn input. The cascade’s expected per-input response time is $\mathbb{E}[R] = \sum_{i=1}^n \bar{e}_i \prod_{j<i} p_j$, and the schedulability question is whether the *worst-case* response time, $R_{wc} = \sum_{i=1}^n e_i^{wc}$, satisfies $R_{wc} \leq D$ for a deadline D . Baruah et al. [2] give an optimal-synthesis algorithm for cascade ordering when both $\bar{e}_i$ and p_i are known, treating the cascade as a static structure: whether to invoke C_{i+1} after an IDK at C_i is fixed at design time.

B. Static Threshold Skipping (Nguyen et al., 2024)

Nguyen, Cheng, and Carroll [3] observed that on ImageNet-V2 TopImages, when ResNet-18 returns a top-1 softmax probability below 0.3, ResNet-34 *itself outputs IDK* in more than 92% of cases. Since time is only saved by skipping B when B would output IDK, their dynamic cascade skips ResNet-34 whenever ResNet-18’s confidence falls below 0.3, jumping directly to ResNet-152; they report 11.5–17.0% average latency reduction across the three ImageNet-V2 subsets (13.8% on Matched-Frequency, GTX 1070 Ti) with no measurable accuracy loss.

C. Random Forest Skipping (Katikaneni et al., 2025)

Katikaneni, Cheng, and Carroll [4] replaced the single threshold with a Random Forest classifier (`n_estimators=50`, `max_depth=4`, `class_weight='balanced'`) trained on three features of the ResNet-18 softmax output $\mathbf{s}$ (with order statistics $s_{(1)} \geq s_{(2)} \geq \dots$): **confidence** $\text{conf} = s_{(1)}$, the top-1 probability; **entropy** $-\sum_i s_i \log s_i$; and **margin** $s_{(1)} - s_{(2)}$, the top-1/top-2 gap. The label is whether ResNet-34 will itself output IDK. They train on the A -abstained samples of Matched-Frequency and TopImages and test on Threshold-0.7. On a Jetson Nano 4 GB embedded target (CPU mode), Random Forest skipping beat both the no-skip and static-threshold cascades on all three ImageNet-V2 subsets; on a desktop-class CPU the static threshold retained an edge on one subset.

III. APPROACH

A. Cascade Architecture

We use the canonical three-stage cascade from [3], [4], $A = \text{ResNet-18} \rightarrow B = \text{ResNet-34} \rightarrow C = \text{ResNet-152}$, all loaded with default `torchvision` pretrained weights (IMAGENET1K_V1). A classifier C_i emits $\perp$ (IDK) when its top-1 softmax probability is below an accuracy threshold $\tau_{\text{acc}} = 0.65$, matching the Nguyen et al. setup; the final classifier C never abstains. The control variable $\text{skip}(\mathbf{x}) \in \{0, 1\}$ is evaluated only when A returns $\perp$: if $\text{skip} = 1$ the cascade jumps directly to C and commits, otherwise it runs B and falls through to C only if B also abstains.

B. Skip-Decision Variants

We compare four implementations of $\text{skip}(\cdot)$:

- 1) **Baseline**: $\text{skip}(\cdot) \equiv 0$. Always run B after A abstains.
- 2) **Static threshold** [3]: $\text{skip}(\mathbf{x}) = 1$ iff $\text{conf}(A) < 0.3$.
- 3) **Random Forest** (re-implementation of [4] with the same hyperparameters): a 50-tree, depth-4 RF classifier on $\{\text{conf}, \text{entropy}, \text{margin}\}$, trained with `class_weight='balanced'` and `min_samples_leaf=40`.
- 4) **Logistic (ours)**: a logistic regression on two features $\{\text{conf}, \text{margin}\}$ only.

C. Logistic Skip Rule

With $\text{conf}(A)$ and $\text{margin}(A)$ as defined in Sec. II-C, the skip predictor is

$$P(\text{skip} = 1 \mid \mathbf{x}) = \sigma(w_0 + w_1 \text{conf}(A) + w_2 \text{margin}(A)),$$

with $\sigma(z) = (1 + e^{-z})^{-1}$. The training label is $y = 1$ when B itself would abstain, i.e. $\max \mathbf{s}_B < \tau_{\text{acc}}$. This is the predictor target of both prior papers [3], [4]: time is only saved by skipping B when B would output IDK; if B would commit, running it is the cheapest path to a committed answer ($t_B < t_C$). Both learned predictors are trained on this label. We fit on the subset of images where A returned $\perp$ (the only inputs where skip is consulted), drawn from the two training subsets under the cross-dataset protocol of Section IV, and apply `class_weight='balanced'` to handle the asymmetric prior ($y = 1$ for 66.9% of A -IDK training images). At inference, the cascade skips B when $P(\text{skip} = 1 \mid \mathbf{x}) > 0.5$; evaluating the predictor reduces to two multiplications, two additions, and one sigmoid.

IV. EXPERIMENTAL SETUP

A. Datasets and Cross-Dataset Protocol

We use all three ImageNet-V2 subsets [5] (10 000 images each, 1 000 classes): *Matched-Frequency*, *TopImages*, and *Threshold-0.7*, obtained as the original per-variant release files of Recht et al. via the Hugging Face Hub, with the variant identity of each split verified against canonical standalone top-1 accuracies. Following the cross-dataset protocol of Katikaneni et al. [4], both learned predictors are trained on the first-stage-IDK samples of Matched-Frequency + TopImages and evaluated entirely on Threshold-0.7: train and test datasets are disjoint, so no within-subset split is needed, the full A -IDK test population is used, and the transfer property that motivated [4] is tested directly.

B. Hardware

All inference is run in CPU mode (`torch.device("cpu")`), matching the GPU-disabled methodology of Katikaneni et al. [4]. CPU is the operative regime: a single ResNet-152 GPU run is already cheap, so the cascade’s value vanishes there. All measurements are taken on an NVIDIA Jetson Orin Nano (ARM Cortex-A78AE, CPU mode), the successor of the Jetson Nano used by [4]: the board runs all model inference (cached softmax outputs and per-image stage timings) as well as the predictor training, evaluation, and per-decision cost profiling.

C. Implementation

Models are loaded from `torchvision.models` with default pretrained weights. Per-image timings are measured with `time.perf_counter()` around the inference call only (preprocessing is excluded). Each measurement is the mean of five trials per image after a 50-image warm-up.

D. Metrics and Evaluation Populations

We report **average latency** (mean wall-clock ms/image over the stated population), **top-1 accuracy** of the cascade’s final committed prediction, **skip rate** (fraction of the population on which B was skipped), and, for the learned predictors, precision/recall/F1 of the skip decision plus measured per-decision cost. Two populations appear in Section V. The threshold sweep uses the *full 10 000-image Matched-Frequency set* (a reproduction of [3]). The learned-predictor comparisons use the *A-IDK population of Threshold-0.7*: the skip rule is consulted nowhere else, and no Threshold-0.7 image appears in training. Numbers from the two populations are not directly comparable (the A-IDK population contains only hard inputs, so its latencies are higher and accuracies lower than full-set values). All point estimates on the test population are reported with bootstrap 95% confidence intervals (10 000 percentile-bootstrap resamples of the test population); comparisons between the Random Forest and the logistic use *paired* bootstrap CIs of the per-image differences.

V. PRELIMINARY RESULTS

All numbers below are measured on the Jetson Orin Nano (CPU mode), with five trials per image after a 50-image warm-up. Table I is computed on the full 10 000-image Matched-Frequency set; Tables II–IV are computed on the Threshold-0.7 A-IDK population under the cross-dataset protocol of Section IV.

A. Threshold Sweep Reproduction

Table I reproduces the threshold-sweep protocol of Nguyen et al. [3]: cascade latency as a function of the ResNet-18 confidence threshold τ used for the skip-to- C rule, on the full 10 000-image Matched-Frequency set. The minimum sits exactly at the $\tau = 0.3$ setting used in the original paper (208.67 ms, with 0.4 statistically indistinguishable at 208.76 ms); it beats the no-skip baseline (216.59 ms) by 3.7%. The gain is smaller than the 13.8% Nguyen et al. report on this subset, reflecting the GPU-vs.-ARM-CPU stage-cost ratio. Accuracy is flat to slightly increasing across the sweep (0.651 to 0.656): below the threshold region B nearly always abstains [3], so replacing B with the more accurate C never hurts the committed prediction.

TABLE I
THRESHOLD SWEEP, FULL MATCHED-FREQUENCY SET (PROTOCOL OF [3])

τ	Latency (ms)	Accuracy	Skip rate
0.0	216.59	0.651	0.000
0.1	215.22	0.651	0.018
0.2	211.41	0.651	0.089
0.3	208.67	0.652	0.181
0.4	208.76	0.653	0.272
0.5	211.20	0.653	0.363
0.6	216.25	0.655	0.448
0.7	219.18	0.656	0.486
0.8	219.18	0.656	0.486
0.9	219.18	0.656	0.486

B. Skip-Predictor Comparison

Table II reports our headline comparison on the Threshold-0.7 A-IDK population (4 051 images), with the predictors trained on Matched-Frequency + TopImages. All three skip rules beat the no-skip baseline (374.86 ms) by about 3%, and the two learned predictors are statistically indistinguishable from each other and from the static threshold on both latency and accuracy — the bootstrap CIs of all three variants overlap substantially. The learned rules do skip far more aggressively (51.2% and 49.3% of A-IDK images vs. 31.5%) at no accuracy cost, but on this transfer target the extra skips do not buy additional latency — consistent with the desktop-CPU edge Katikaneni et al. themselves observed; the stage-cost ratio is the operative variable (Sec. VI). Between the two learned predictors, the paired bootstrap 95% CIs of the per-image RF–logistic differences are $[-2.62, -0.28]$ ms for latency and $[-0.002, +0.002]$ for accuracy: the Random Forest’s latency edge is real but bounded by 2.6 ms — smaller than its own per-decision evaluation cost, measured next.

Point estimates at this scale are fragile: across 10 random 70/30 within-subset splits (Matched-Frequency), Random Forest skip-F1 was 0.680 ± 0.014 vs. 0.704 ± 0.010 for the logistic — the split-to-split spread exceeds the gap between predictors, hence interval estimates throughout.

Table III gives the feature-selection evidence behind the lightweight choice: nested logistic models on {conf}, {conf, margin}, and {conf, margin, entropy}. Under cross-dataset transfer, confidence alone reaches F1 0.676 [0.661, 0.692]; adding margin (0.662) and entropy (0.662) changes nothing outside the confidence intervals. The feature set of [4] thus collapses, under transfer, to its first coordinate — a learned one-feature logistic is exactly a dataset-calibrated confidence threshold, connecting the two prior papers directly.

TABLE II
CASCADE VARIANTS, THRESHOLD-0.7 A-IDK POPULATION (95% CIs)

Variant	Latency (ms)	Accuracy	Skip rate
Baseline (no skip)	374.86 [369.91, 379.82]	0.549 [0.533, 0.564]	0.000
Static $\tau = 0.3$	363.65 [359.30, 368.01]	0.552 [0.537, 0.567]	0.315
Random Forest	363.83 [360.03, 367.63]	0.553 [0.538, 0.568]	0.512
Logistic (ours)	365.28 [361.48, 369.05]	0.553 [0.538, 0.568]	0.493

TABLE III
SKIP-PREDICTOR QUALITY AND FEATURE ABLATION (95% CIs ON F1)

Predictor	Precision	Recall	F1
RF {conf, entropy, margin}	0.778	0.607	0.682 [0.667, 0.697]
Logistic {conf}	0.778	0.598	0.676 [0.661, 0.692]
Logistic {conf, margin} (ours)	0.772	0.580	0.662 [0.647, 0.678]
Logistic {conf, margin, entropy}	0.772	0.579	0.662 [0.647, 0.678]

C. Measured Predictor Overhead

Table IV reports the per-decision cost of each learned predictor, measured at batch size 1 over 1 000 test samples after

warm-up (`time.perf_counter`, on the Jetson Orin Nano): the scikit-learn Random Forest via `predict()`, the logistic in its deployed form (a NumPy dot product plus sigmoid). The median gap is $205\times$ (3.80 ms vs. 19 μ s). Because worst-case behavior is what enters a schedulability bound, we also report the empirical p99 and maximum (6.37 ms vs. 48 μ s). This measures the deployed scikit-learn implementation — a compiled tree ensemble would narrow the gap — but the logistic’s two-multiply evaluation is a floor no 50-tree ensemble reaches. Beyond magnitude, the two predictors differ in *analyzability*: the deployed logistic is branch-free and data-independent (exactly two multiplications, two additions, and one sigmoid on every input), so a WCET bound is immediate, whereas a tree ensemble’s evaluation path is input-dependent and its interpreted implementation resists tight static bounding. The cascade latencies in Table II *exclude* predictor cost; on the A-IDK population, where the predictor is consulted on every image, charging each variant its median per-decision cost adds 3.80 ms to the Random Forest row but only 0.02 ms to the logistic row — *more than the Random Forest’s entire paired latency advantage over the logistic* (≤ 2.6 ms). End-to-end, the cheapest learned rule is the logistic.

TABLE IV
PER-DECISION PREDICTOR COST (BATCH 1, JETSON ORIN NANO)

Predictor	Mean (μ s)	Median (μ s)	p99 (μ s)	Max (μ s)	Relative (median)
Random Forest	3819.3	3803.7	4036.4	6365.5	$205\times$
Logistic (ours)	18.8	18.6	25.4	48.0	$1\times$

a) *Learned coefficients.*: The logistic predictor’s two-feature linear discriminant is, in the trained model,

$$\text{logit } P(\text{skip} \mid \mathbf{x}) = 1.841 - 5.481 \text{ conf}(A) + 1.785 \text{ margin}(A).$$

The dominant term matches intuition: higher confidence sharply reduces skip probability (a confident A suggests an input easy enough for B to commit on). The smaller positive margin coefficient applies a correction conditional on confidence: at fixed top-1 confidence, a larger top-1/top-2 gap slightly increases the predicted probability that B abstains. Confidence dominates — consistent with confidence being the single feature used by the static threshold rule.

VI. DISCUSSION AND FUTURE WORK

Why so few features suffice. On a $K=1000$ -class output, the bulk of entropy lives in the long tail of small probabilities; whether B will abstain is dominated by A ’s top-1 probability. The nested-model ablation (Table III) makes this concrete: under cross-dataset transfer, neither margin nor entropy improves on confidence alone outside the bootstrap CIs. The interesting corollary is that a one-feature logistic *is* a confidence threshold whose cut point is learned from training data — so the practical gap between [3] and [4] is not the classifier family but whether the threshold is calibrated per deployment.

When does skipping pay? A skip saves t_B when B would abstain and wastes $t_C - t_B$ when B would have committed, so a skip rule breaks even at precision $p^* = 1 - t_B/t_C$. On our board ($t_B \approx 97$ ms, $t_C \approx 339$ ms), $p^* \approx 0.71$, and both learned predictors operate at precision ≈ 0.77 (Table III) — barely above break-even, which is why their aggressive skipping (Table II) yields only $\sim 3\%$ latency gains. The stage-cost ratio, not the predictor, is the operative variable: larger t_C/t_B raises p^* , while cheap final stages make even mediocre skip rules profitable.

Real-time implications. The skip predictor’s own WCET enters the schedulability budget. Our measured per-decision cost gap is $205\times$ at the median and $133\times$ at the empirical maximum (Table IV), and the logistic’s branch-free, data-independent evaluation admits an immediate WCET bound where the tree ensemble’s input-dependent path does not; in Baruah’s framing [2], where each stage’s WCET is charged against the response-time bound, the near-zero-cost predictor is preferable whenever cascade accuracy and latency are comparable — which Table II shows they are, before the Random Forest is even charged its own overhead.

Future work. Confidence is well-defined for any softmax classifier, so the rule is modality-agnostic (e.g., audio pipelines [6]); natural follow-ons are folding the predictor’s WCET and p^* into Baruah’s response-time bound and multi-stage skip rules for longer cascades.

VII. CONCLUSION

Under Katikaneni et al.’s cross-dataset protocol, measured end to end on a Jetson Orin Nano, a two-feature logistic skip rule matches the Random Forest’s cascade accuracy and concedes at most 2.6 ms of cascade latency — less than the Random Forest’s own 3.8 ms median per-decision cost, which the logistic undercuts $205\times$ with a branch-free evaluation whose WCET is immediate to bound. The feature ablation shows confidence alone suffices, making the learned rule a dataset-calibrated threshold; the break-even precision $p^* = 1 - t_B/t_C$ explains when any skip rule pays. Folding both into an end-to-end schedulability analysis is the immediate next step.

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